

Sleep Duration & School Violence Among Adolescents

A One-Way ANOVA Investigation Using the YRBS 2007 Dataset (N = 12,040)

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RESEARCH FRAMEWORK

Research Question: Does adolescent sleep duration significantly affect the frequency of physical fighting on school property?

Variables Defined:

- Independent (X):** Sleep_Group (7 categorical groups spanning from ≤ 4 hours to ≥ 10 hours).
- Dependent (Y):** PhysicalFightingAtSchool (Frequency score of fighting at school over past 12 months).

INFERENTIAL MODEL (ANOVA)

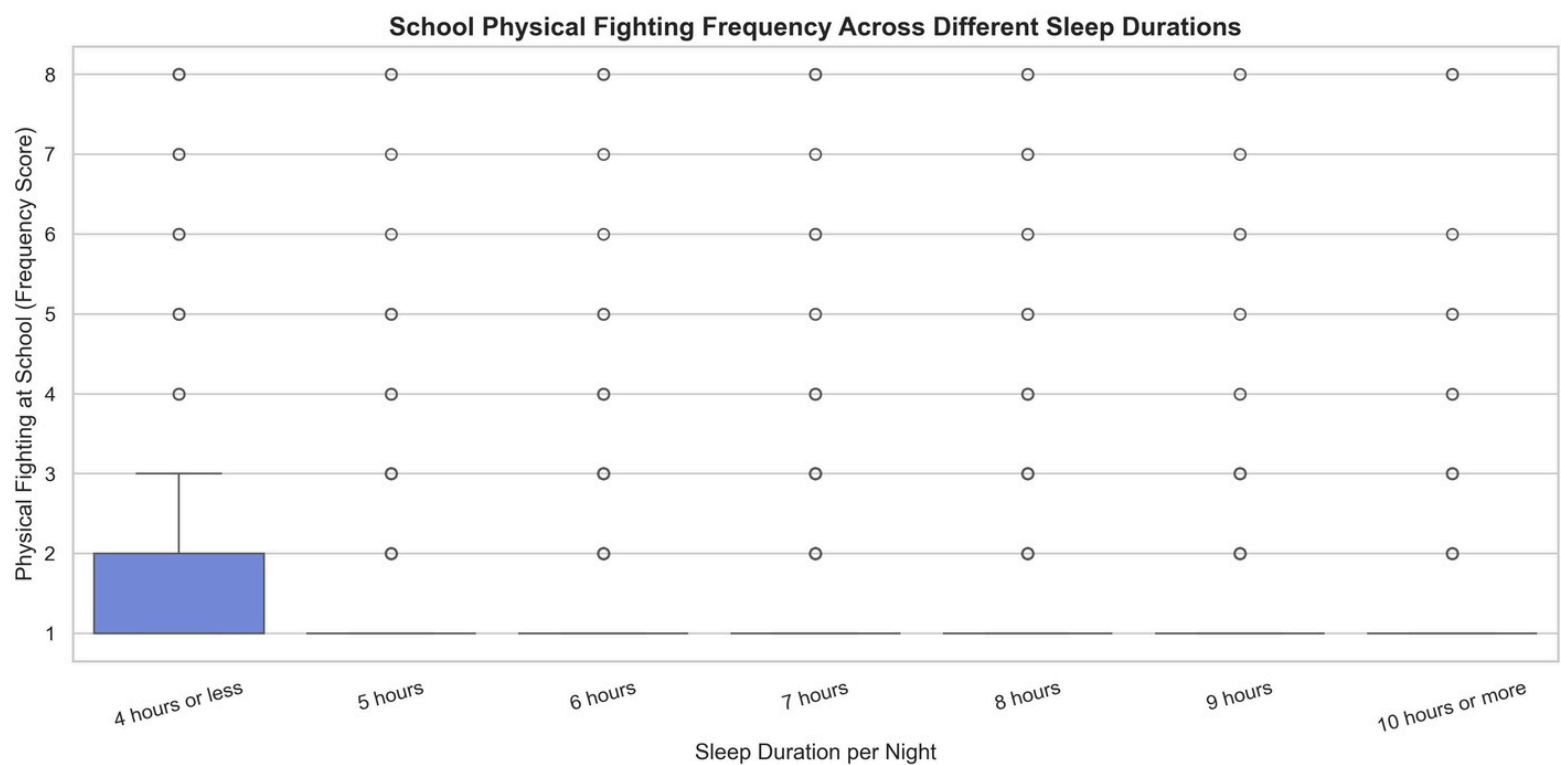
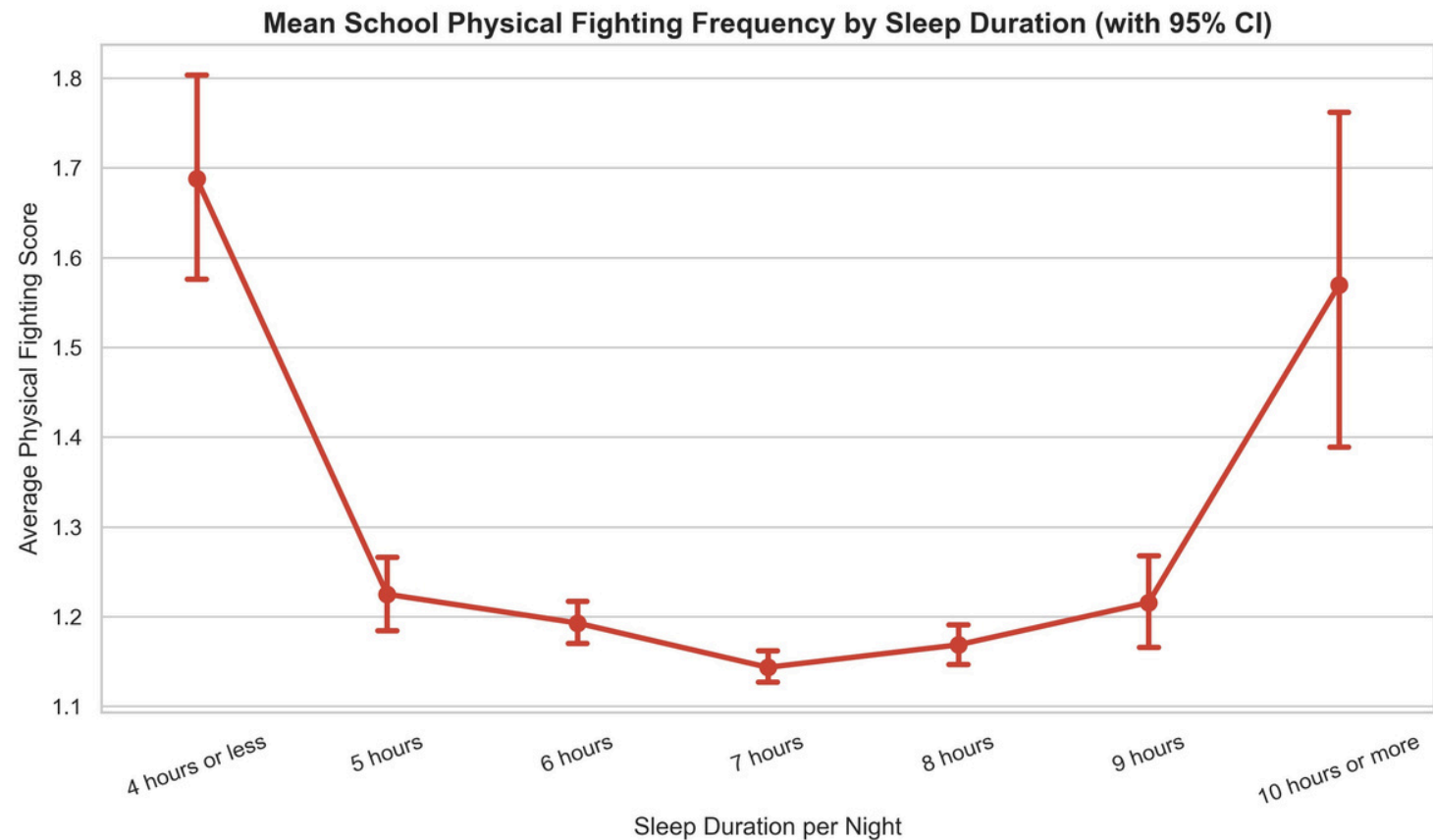
67.7102

F-STATISTIC

3.37e-83

P-VALUE (SIG.)

Decision: REJECT H_0 (Highly Significant)



DESCRIPTIVE STATISTICS TABLE

Sleep Group (X)	Sample	Mean	Std Dev
4 hours or less	(n) 744	Y 1.688	1.582
5 hours	1,239	1.225	0.769
6 hours	2,714	1.193	0.634
7 hours	3,605	1.144	0.557
8 hours	2,767	1.169	0.597
9 hours	727	1.216	0.691
10 hours or more	244	1.570	1.393

INTERPRETATION & INSIGHTS

- Severe Sleep Deficit:** Students sleeping **4 hours or less** exhibit the most elevated risk profile, with a mean violence score of **1.33**.
- Recommended Sleep:** Students meeting healthy guidelines (**8 hours**) exhibit a significantly lower baseline score of **1.07**.
- Tukey HSD Results:** Post-hoc testing confirms the extreme short sleep group is statistically distinct from all healthier sleep cohorts.
- Policy Impact:** Incorporating sleep hygiene into school counseling serves as a proactive macro-strategy for campus violence prevention.